# Harshit Uppala

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LinkedIn | GitHub | Portfolio
Available full-time December 15, 2026; authorized to work in Canada without sponsorship.

## Profile

Computer Science candidate focused on Python backend engineering and applied AI integration. Project work includes FastAPI services, durable processing workflows, PostgreSQL persistence, a React/TypeScript interface, and independent evaluation. Brings a Business Administration minor and AWS Cloud Practitioner certification.

## Technical Skills

**Languages:** Python, TypeScript, JavaScript, SQL
**AI applications:** LangGraph, structured outputs, RAG, embeddings, BM25, pgvector, evaluation
**Backend and delivery:** FastAPI, React, PostgreSQL, REST APIs, Git, pytest, OpenTelemetry; AWS fundamentals

## Selected Engineering Projects

**ATLAS — Enterprise Requirements Intelligence** *Active development*
*Python, FastAPI, LangGraph, React/TypeScript, PostgreSQL/pgvector, Amazon Bedrock*

- Developed FastAPI endpoints for bounded PDF/DOCX ingestion, queued document analysis, source retrieval, and versioned human review, connecting AI-oriented processing to an application workflow.
- Implemented database-backed jobs with idempotent admission, fenced leases, cancellation, deadlines, and atomic publication; tested recovery and stale-worker behavior.
- Built a React/TypeScript evidence interface backed by tenant-scoped PostgreSQL persistence and exact source-citation validation; executed local restricted-login database integration tests.
- Integrated CPU embeddings, lexical retrieval, and reciprocal-rank fusion; developed an opt-in Bedrock structured-output boundary with validation and budget controls.

**CHAOSLAB — AI Workflow Evaluation** *Active development*
*Python, pytest, independent adapters, reproducible reports*

- Developed an independent Python evaluation runner with authored fixtures, adapter contracts, and reproducible reports to test analysis quality and failure behavior.
- Added regression scenarios for interrupted jobs, tenant boundaries, malformed inputs, and forged publication metadata; documented architecture decisions and remaining limitations.
- Project verification records 247 passing default backend tests and 18 local PostgreSQL integration groups. Model-quality benchmarks and cloud application deployment remain separate validation work.

## Education

**Memorial University of Newfoundland** Expected December 2026
B.Sc. Computer Science; Minor in Business Administration

## Certifications

**AWS Certified Cloud Practitioner** — earned, 2026
AWS Certified Solutions Architect – Associate — exam scheduled
Claude Certified Architect – Foundations — planned (target November 2026)